



Kendall Johnson (Attempt 1 of 1)

Test Information

QUESTION 1: CALCULATED NUMERIC

5

out of 5 points

Consider the control of a process with transfer function

$$G_P(s) = \frac{0.5e^{-5s}}{(20s+1)(5s+1)}$$



For a proportional controller with gain K_C at what value of the gain the process begins to become unstable. Round your numerical answer into **four decimal digits**.

Given Answer: 11.6619

Correct Answer: 11.9598

Answer range +/- 0.0001 (11.95970 - 11.95990)

QUESTION 2: CALCULATED NUMERIC

0

out of 5 points

Determine the Open Stem Travel (%Valve Open) of a 4in inline control valve in an 8in NPS Schedule 40 pipeline, with inherent characteristic given in **Table 1** in the **CHE Calculator** that is used to regulate the flow of liquid propane. The service conditions are: Flowing temperature $T=50^\circ\text{F}$, volumetric design flow rate $Q=250$ gpm, upstream pressure $P_1=350$ psia and downstream pressure $P_2=300$ psia. Express your result in percent travel (for instance if you answer is 25.3% enter the value 25). Assume that the **Peng-Robinson** Equation of State can be used for the calculation of the thermal properties of propane and the **Rackett** equation for the liquid density. **Round your answer to a whole number**.



Given Answer: 58

Correct Answer: 45

Answer range +/- 0 (45 - 45)

QUESTION 3: CALCULATED NUMERIC

0

out of 5 points

Using the Bode stability analysis determine the minimum value of the reset time τ_I for stability of a **PI** controller with gain $K_C = 0.3$ applied to a process with transfer function

$$G_P(s) = \frac{e^{-0.5s}}{s+1}$$



Round your numerical answer into **four decimal digits**.

Given Answer: 0.2245

Correct Answer: 0.1118

QUESTION 4: MULTIPLE ANSWER

10

out of 10 points

Consider the process transfer function: $G_P(s) = \frac{1}{-5s+1}$. Assume that the controller gain K_c is positive. Using the Routh–Hurwitz criterion, determine which of the following controllers can stabilize the loop. Select all that apply.



Given  Proportional Derivative (PD): $G_c(s) = K_c(1 + \tau_D s)$

Answers:

- Proportional Integral Derivative (PID): $G_c(s) = K_c \left(1 + \frac{1}{\tau_I s} + \tau_D s \right)$

Correct Proportional Derivative (PD): $G_c(s) = K_c(1 + \tau_D s)$

Answers:

- Proportional Integral Derivative (PID): $G_c(s) = K_c \left(1 + \frac{1}{\tau_I s} + \tau_D s \right)$

QUESTION 5: CALCULATED NUMERIC

5

out of 5 points

A controller sends an 12mA analog signal to an I/P converter. The pressure signal from the I/P converter is sent to the control valve. What is the valve opening as a percentage of the total travel, assuming total travel corresponds to 100% (fully open valve)



Given Answer:  50

Correct Answer: 50

Answer range +/- 0 (50 - 50)

QUESTION 6: SHORT ANSWER

10

out of 10 points

Consider the process transfer function: $G_P(s) = \frac{1}{-5s+1}$ Assume that the controller gain K_c is positive. Select a controller that stabilizes the closed loop response and derive the stability conditions on the controller parameters. Write your answer in the textbox below.

Given A PD controller stabilizes the system.

Answer:

$$G_C(s) = K_C(1 + \tau_D s)$$

$$1 + G_C(s) G_P(s) = 0$$

$$s = -\frac{1 + K_C}{-5 + K_C \tau_D}$$

$$K_C \tau_D > 5, K_C > 0$$

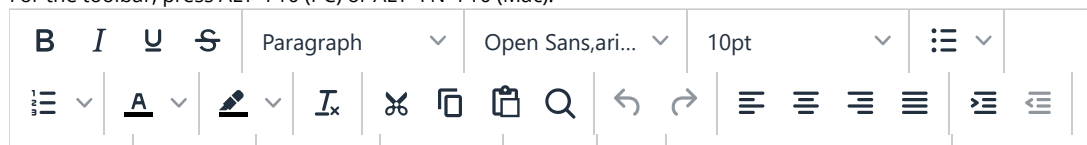
Correct [None]

Answer:



Response For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

Feedback:



0

Consider a first order process with time delay $G_P(s) = \frac{1.2e^{-3.7s}}{13.7s+1}$. The process is regulated with a PI controller with Gain $K_C = 0.45 K_u$ and Reset Time $\tau_I = 0.83 P_u$. Perform a simulation for a unit size step change in the **Set Point** and calculate the overshoot and the settling time. Enter

1. The additional time delay the loop can tolerate before it becomes unstable
2. The overshoot to the step change in the set point
3. The settling time after the step change in the set point

Given Solving for Ku and Pu (using the initial guesses and the Newton Method) through the Bode Stability, Kc =
Answer: 2.1389, TI=12.0204.

1. The loop grows unstable but still in relative consistent osillations for a lag time of 8, entirely unstable at a lag time of 9.
2. Approx. 25% (for the inital condition)
3. Approx. 43.8 (for the initial condition)



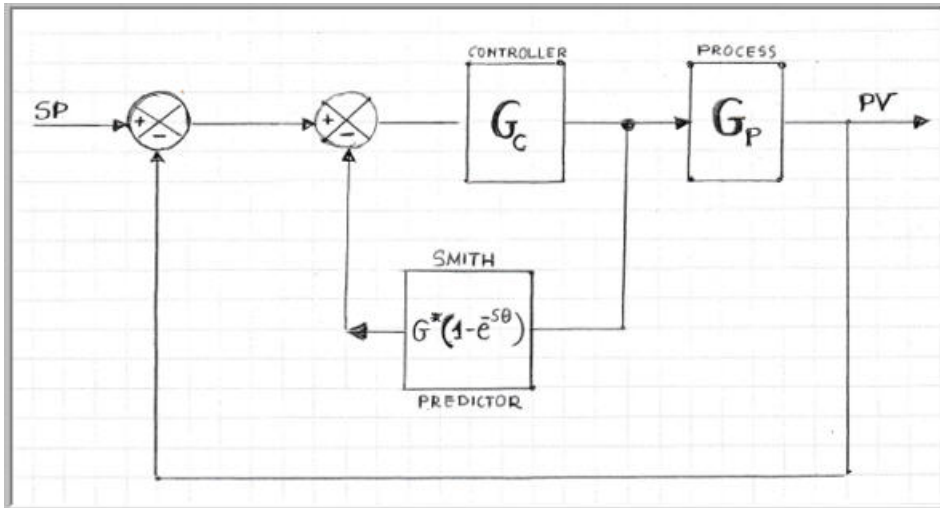
Correct
Answer:

[None]

Response For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

[illegible]

Time delays frequently occur in the process industries because of the presence of transportation lags, recycle loops and sensor inertia. Time delays result in reduction of the controller gain in order to maintain stability and as a result make the controller performance sluggish. To improve the performance of loops with time delays, advanced control strategies have been developed over the years. One such strategy is known as the **Smith predictor** and it consists of an ordinary feedback loop plus an inner loop that introduces an extra term directly into the feedback path. A control block diagram of the Smith predictor controller is given below:



where $G_P(s) = G_P^*(s) e^{-s\theta}$ and $G_P^*(s)$ is the process transfer function free of the time delay. Derive the



closed-loop transfer function $G_{CL}(s) = \frac{PV}{SP}$ and enter you final formula in the textbox below:

Given $G_c(s) G_P^*(s) e^{-s\theta}$
 Answer: $\frac{G_c(s) G_P^*(s) e^{-s\theta}}{1 + G_c(s) G_P^*(s)}$

Correct [None]
Answer:

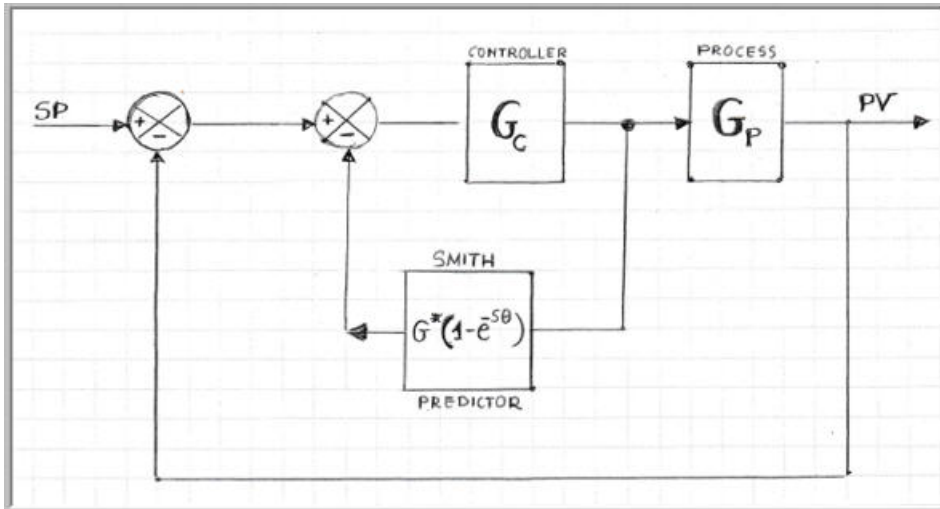
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Feedback:

QUESTION 9: SHORT ANSWER

5 out of 20 points

Time delays frequently occur in the process industries because of the presence of transportation lags, recycle loops and sensor inertia. Time delays result in reduction of the controller gain in order to maintain stability and as a result make the controller performance sluggish. To improve the performance of loops with time delays, advanced control strategies have been developed over the years. One such strategy is known as the **Smith predictor** and it consists of an ordinary feedback loop plus an inner loop that introduces an extra term directly into the feedback path. A control block diagram of the Smith predictor controller is given below:



where $G_P(s) = G_P^*(s) e^{-s\theta}$ and $G_P^*(s)$ is the process transfer function free of the time delay. Consider the



following first order process $G_P(s) = \frac{1.3e^{-3.7s}}{13.7s+1}$. This chemical process is controlled with a PI feedback

controller **with the Smith-Predictor in the loop**. The delay time in the Smith-Predictor is set at $\theta = 3.7$.

Perform a simulation for a unit step change in the set point and calculate the **overshoot** and the **settling time**. Enter your numerical results separated by comma in the textbox below:

Given 76,178

Answer:

Correct [None]

Answer:

Response For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

Feedback:

[illegible]

out of 10 points

FEEDBACK AND NOTES FOR ATTEMPT